SERK Construction OS

Software Requirements Specification

Enterprise Construction ERP | Project Controls | Procurement and Subcontracts | Finance | Field and Offline | Governed AI

Based on: SERK Construction repository baseline (platform 0038) and enterprise target state

Document version 1.0 - DRAFT FOR AGREEMENT
Date: 6 November 2026
Structure follows ISO/IEC/IEEE 29148 (requirements engineering)
Deployment: multi-tenant SaaS, dedicated, in-country or on-premises

Not legal, tax, accounting or regulatory advice. Applicability must be confirmed by qualified professionals.

Document Control

Field	Value
Document title	SERK Construction OS: Software Requirements Specification (SRS)
Version / status	1.0 / DRAFT, issued for review and formal agreement
Date	6 November 2026
Product	SERK Construction OS: enterprise construction ERP (finance, project controls, procurement, subcontracts, inventory, equipment, workforce, contracts, field, documents, QHSE, intelligence)
Source baseline	Repository *Serk-construction* at commit 5f94bc3 (platform 0038, 38 migrations, ADRs 0001-0017, recorded run of 378 tests: 0 failed, 0 skipped), extended to an enterprise target state
Prepared for	Product owner, engineering, finance, commercial, project controls, operations, HSE/QA, security, compliance and QA stakeholders
Intended use	Contractual and engineering baseline: what the system shall do and the qualities it shall have. How it is built is governed by the ADRs and the architecture documents.

Revision history

Version	Date	Author	Change
0.1	2026-11-06	Drafting	Derived from the implemented baseline and enterprise market requirements
1.0	2026-11-06	Drafting	Complete first issue for stakeholder review

How to read and use this document

- Every requirement has a unique ID, a MoSCoW priority (**Must / Should / Could**), a verification method (**Test / Inspection / Demo / Analysis**) and a **Baseline** status.
- **Baseline** states how far the current codebase already meets the requirement: `Exists` (implemented and covered by automated tests), `Partial` (some of it exists) or `New` (to be built). It is engineering evidence, not acceptance: every requirement is still verified as written.
- **shall** = mandatory; **should** = recommended, may be deferred by agreement; **may** = optional.
- Items in Appendix E (Open Decisions) need an owner and a decision before the related requirements can be frozen. After sign-off (Section 15), changes follow Section 14.

Important notice. This SRS defines requirements; it is not a certification, a legal or tax opinion, or an accounting policy. Tax, payroll, labour, data-protection and IFRS treatment must be confirmed by qualified professionals before live use. The current release decision recorded in the repository is **NO-GO** for production until the named gates and human sign-offs exist.

Table of Contents

Document Control 2

1. Introduction 5

- 1.1 Purpose 5
- 1.2 Scope 5
- 1.3 Product vision and quality pillars 5
- 1.4 Definitions, acronyms and abbreviations 5
- 1.5 References 6
- 1.6 Document organisation 6

2. Overall Description 7

- 2.1 Product perspective 7
- 2.2 Logical modules 7
- 2.3 User classes and characteristics 8
- 2.4 Operating environment 8
- 2.5 Design and implementation constraints 8
- 2.6 Market position and differentiation 9
- 2.7 Assumptions and dependencies 9

3. Core Business Rules and Invariants 10

- 3.1 Authoritative write path 10
- 3.2 Controlled document lifecycle (reference) 10

4. External Interface Requirements 11

- 4.1 User interfaces 11
- 4.2 Hardware interfaces 11
- 4.3 Software and API interfaces 11
- 4.4 Communications interfaces 11

5. Principal Use Cases 12

6. Functional Requirements 13

- 6.1 Identity, Tenancy and Access Control 13
- 6.2 Organisation and Master Data 14
- 6.3 Business Development, Estimating and Tendering 14
- 6.4 Project Controls: Budget, Commitment, Cost and Forecast 15
- 6.5 Planning and Scheduling 16
- 6.6 Progress Measurement and Earned Value 16
- 6.7 Procurement and Supply Chain 16
- 6.8 Subcontract Management 17
- 6.9 Inventory and Materials 17
- 6.10 Equipment and Fleet 18
- 6.11 Workforce, Time and Payroll 18
- 6.12 Contracts, Revenue and Claims 19
- 6.13 Finance and Accounting 20
- 6.14 Treasury and Cash Management 20
- 6.15 Site Operations and Mobile 21
- 6.16 Document Control and BIM 21
- 6.17 Quality Management 22
- 6.18 Health, Safety and Environment 22
- 6.19 Workflow, Approvals and Automation 23
- 6.20 Reporting, Dashboards and Analytics 23

6.21 Embedded Intelligence and AI 23
6.22 Audit, Compliance and Traceability 24
6.23 Collaboration, Notifications and Portals 24
6.24 Integration and Interoperability 25
6.25 Administration, Configuration and Localisation 25

7. Non-Functional Requirements 27

7.1 Performance and Capacity 27
7.2 Availability, Reliability and Recovery 27
7.3 Security 27
7.4 Privacy and Regulatory Compliance 28
7.5 Usability, Accessibility and Localisation 28
7.6 Maintainability, Portability and Operations 29
7.7 Data Integrity 29

8. Data Requirements 31

8.1 Conceptual entities 31
8.2 Data retention, migration and quality 31

9. Verification and Release Gates 33

9.1 Acceptance criteria summary 33
9.2 Release plan 33

10. Client Technology Position 34

11. Assumptions and Risks 35

12. Quality Assurance Approach 35

13. Commercial and Support Requirements 35

14. Change Control 36

15. Agreement and Sign-off 36

Appendix A - Requirement Counts by Priority and Baseline 37

Appendix B - Traceability: Baseline Architecture to SRS 39

Appendix C - Reference Test Case Catalogue 39

Appendix D - Role-Capability Matrix (initial proposal) 41

Appendix E - Open Decisions for Agreement 42

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of SERK Construction OS, an enterprise resource planning platform for construction contractors. It covers the whole contractor lifecycle: winning work, controlling project cost and time, buying and subcontracting, running sites, paying people and suppliers, billing clients, and closing the books. It is the single reference against which design, implementation, testing and acceptance are measured, and it sets the scope needed to compete with established international construction and ERP suites.

1.2 Scope

In scope: multi-company, multi-project construction operations: organisation and master data; business development, estimating and tendering; project controls (budget, commitment, cost, forecast, WIP); planning, scheduling and earned value; procurement and import; subcontract management; inventory; equipment and fleet; workforce, time and payroll; client contracts, variations, claims and revenue; finance, tax, fixed assets and consolidation; treasury; site operations and an offline mobile application; document control and BIM; quality; health, safety and environment; workflow and approvals; reporting and analytics; governed AI; audit; portals; integration and administration.

Out of scope for release 1.x (unless added by change control): design authoring (CAD/BIM modelling), real-estate sales and property management, manufacturing (precast plant MRP), full recruitment and talent management, e-commerce, and direct movement of money (the system issues instructions and reconciles; banks settle).

1.3 Product vision and quality pillars

Pillar	Meaning for requirements
CONTROLLED	Every financial and commercial decision is made by an authorised human, checked by another, attested cryptographically and re-verified by the database. Controls hold even if the application is bypassed.
INTEGRATED	One system of record from estimate to final account: budget, commitment, actual, forecast and revenue come from the same documents and the same ledger, with no reconciliation between separate products.
FIELD-READY	Works on intermittent connectivity and on phones; field data is captured where work happens and synchronised safely.
INTELLIGENT	Analytics and AI explain, predict and draft from permission-filtered facts with sources; they never decide.
LOCAL AND OPEN	Built for Ethiopia and East Africa (languages, calendar, currency, tax, payroll, contract forms) and open through documented APIs, in any deployment model.

1.4 Definitions, acronyms and abbreviations

Term	Definition
ABAC / RBAC	Attribute-based / role-based access control
AP / AR / GL	Accounts payable / accounts receivable / general ledger
Attestation	A signed statement, issued by the isolated signer for an authenticated human session, that the database verifies before accepting a decision
BOQ	Bill of quantities
CDE	Common data environment (ISO 19650)
CPM / EVM	Critical path method / earned value management (PV, EV, AC, SPI, CPI, EAC, ETC, VAC)

Term	Definition
EOT	Extension of time
FIFO / WAC	First in, first out / weighted average cost
GRN / PO / PR / RFQ	Goods receipt note / purchase order / purchase requisition / request for quotation
IPC	Interim payment certificate
ITP / NCR	Inspection and test plan / non-conformance report
Maker / checker	The person who prepares a document and a different person who approves it
MFA / step-up	Multi-factor authentication / a fresh MFA required for a specific sensitive action
Outbox	Events written in the same transaction as the business change and delivered after commit
RLS	PostgreSQL row-level security
RPO / RTO	Recovery point objective / recovery time objective
SoD	Segregation of duties
Tenant	A construction company using the product, isolated from all others
WBS	Work breakdown structure
WIP	Work in progress (over/under-billing schedule)

1.5 References

- SERK Construction repository: README, docs/architecture (MODULES, DATA_FLOW, THREAT_MODEL), ADRs 0001-0017, docs/release (READINESS, GO_NO_GO), docs/BACKLOG.md, api/openapi.yaml.
- IFRS (IAS 16, IAS 21, IFRS 15, IFRS 10/11); Ethiopian tax, labour and pension legislation as confirmed by advisers.
- Ethiopia Personal Data Protection Proclamation No. 1321/2024; GDPR where applicable.
- ISO 19650 (information management using BIM); IFC and BCF (buildingSMART); FIDIC conditions of contract; Ethiopian public-procurement standard bidding documents.
- ISO 20022 (payments); ISO 15143-3 (telematics); OWASP ASVS 4; WCAG 2.1; ISO/IEC 27001; SOC 2; ISO/IEC/IEEE 29148 (requirements engineering).

1.6 Document organisation

Section 2 describes the product context and market position; Section 3 the business rules and invariants; Section 4 the interfaces; Section 5 the principal use cases; Section 6 the functional requirements; Section 7 the non-functional requirements; Sections 8 and 9 data, verification and release; Section 10 the client technology position; Sections 11 to 15 risks, quality assurance, change control and sign-off. The appendices give counts, traceability, the test catalogue, the role matrix and open decisions.

2. Overall Description

2.1 Product perspective

SERK Construction OS is a multi-tenant **modular monolith** written in Go on **PostgreSQL**, the authoritative store and last line of defence. Business commands pass through a kernel that checks permission, project scope and segregation of duties, then commit the change, its audit record and its outbox event in one transaction. An isolated **approval signer** holds the attestation key. Separate processes deliver outbox events (relay), run migrations and serve the web console. Users reach the system through a browser console, an offline-capable mobile application and external portals over HTTPS. Integrations use a versioned REST API and signed webhooks.

2.2 Logical modules

Module	SRS section
Identity, Tenancy and Access Control	6.1
Organisation and Master Data	6.2
Business Development, Estimating and Tendering	6.3
Project Controls: Budget, Commitment, Cost and Forecast	6.4
Planning and Scheduling	6.5
Progress Measurement and Earned Value	6.6
Procurement and Supply Chain	6.7
Subcontract Management	6.8
Inventory and Materials	6.9
Equipment and Fleet	6.10
Workforce, Time and Payroll	6.11
Contracts, Revenue and Claims	6.12
Finance and Accounting	6.13
Treasury and Cash Management	6.14
Site Operations and Mobile	6.15
Document Control and BIM	6.16
Quality Management	6.17
Health, Safety and Environment	6.18
Workflow, Approvals and Automation	6.19
Reporting, Dashboards and Analytics	6.20
Embedded Intelligence and AI	6.21
Audit, Compliance and Traceability	6.22
Collaboration, Notifications and Portals	6.23
Integration and Interoperability	6.24
Administration, Configuration and Localisation	6.25

2.3 User classes and characteristics

Actor	Primary goals	Key permissions
Executive / Owner / CFO	Portfolio performance, cash, risk	Group-wide read, high-value approvals, policy approval, period reopen
Business development / Estimator	Win profitable work	Opportunities, estimates, BOQ drafting
Commercial manager / Quantity surveyor	Protect contract value	BOQ approval, measurement, IPC, variations, claims, subcontract certification
Project manager / Project controls / Planner	Deliver on time and budget	Budget revisions, requisition approval, schedules, progress, forecasts
Site engineer / Foreman	Execute work safely	Requisitions, receipt confirmation, diaries, timesheets, inspections, incidents (mobile, offline)
Procurement officer / manager	Buy right, on time	Suppliers, RFQs, awards, purchase orders, subcontracts
Storekeeper / Inventory controller	Account for every item	Receipts, issues, transfers, counts and their approval
Equipment manager	Keep plant productive	Register, usage, maintenance, fuel, transfers
HR / Payroll officer	Pay people correctly	Employees, timesheets, payroll, statutory returns
Accountant / AP / AR / Controller / Finance manager	Correct, timely books	Journals, subledgers, reconciliations, close, consolidation
Treasury officer / Bank statement officer	Liquidity and safe payments	Payment preparation and release; independent statement import
QA/QC and HSE officers	Quality and zero harm	ITPs, NCRs, permits, incidents, inspections
Auditor (internal / external)	Independent assurance	Read-only data and audit log, time-boxed
Process owner / Workflow admin / Security admin	Govern the system	Workflows, automation, settings, sessions, devices; no financial authority
External: client, consultant, supplier, subcontractor	Collaborate in their own scope	Portal access to their own documents and actions

2.4 Operating environment

- **Server:** Linux containers; Go services (API, signer, relay, migrator, workers); PostgreSQL 16+ with point-in-time recovery; S3-compatible object storage; secret vault and HSM/KMS; TLS edge with WAF.
- **Deployment models:** multi-tenant SaaS, dedicated single-tenant, in-country hosting, and on-premises including air-gapped.
- **Clients:** current desktop browsers; tablets and phones; Android (priority) and iOS mobile app; intermittent connectivity on site must be expected.

2.5 Design and implementation constraints

- The server is the authority for every amount, status and permission; the client holds none.
- Controls are enforced twice: in the kernel and by PostgreSQL (constraints, triggers, RLS, attestation checks).
- Money is exact decimal; migrations are additive; posted records are never edited or deleted.
- No AI, automation or service principal holds decision authority.
- Payments are instructions only; settlement comes from bank evidence.

2.6 Market position and differentiation

Contractors today typically combine a finance ERP (for example SAP S/4HANA, Oracle Fusion, Microsoft Dynamics, Sage) with separate project-controls, scheduling, field and document products (for example Oracle Primavera and Aconex, Procore, Autodesk Construction Cloud, Trimble Viewpoint, InEight, RIB). SERK competes by offering that scope as one controlled system with strong local fit. The comparison below is indicative and must be refreshed by product marketing before external use.

Capability	Typical market offering	SERK target position
Financial control depth	Strong (SAP, Oracle Fusion); weaker in field-first platforms, which rely on an ERP integration	One ledger; controls enforced in the database as well as the app
Project controls (budget, commitment, EVM, forecast)	Strong in specialists (InEight, Oracle Primavera/Unifier, RIB)	Native and linked to the ledger, so no reconciliation between systems
Field and offline	Strong in Procore and Autodesk Construction Cloud; heavy in classic ERPs	Offline-first with idempotent sync, designed for low connectivity
Approvals and SoD	Configurable; enforced in the application layer	Cryptographic attestations re-verified by the database; humans only
AI	Rapidly growing assistants	Advisory only, permission-filtered, computed numbers, auditable
Local fit (Ethiopia / East Africa)	Partner localisation, typically limited	Amharic and other languages, ETB, Ethiopian calendar and fiscal year, local tax, payroll and contract forms
Deployment	Mostly cloud-only SaaS (field platforms) or heavy on-premises (classic ERP)	SaaS, dedicated, in-country or air-gapped on-premises from one codebase
Total cost of ownership	High licence and implementation cost	Tiered pricing; guided onboarding; templates

2.7 Assumptions and dependencies

- Tenants provide or adopt an identity provider for SSO, or use the built-in one with MFA.
- Qualified advisers confirm tax, payroll, labour, IFRS and data-protection treatment (Appendix E).
- Banks and mobile-money providers offer files or APIs; where they do not, statement import remains the settlement evidence.
- Customers supply legacy data for migration and accept a reconciliation sign-off.

3. Core Business Rules and Invariants

These invariants govern every requirement. Each has a release gate (Section 9); failure of any blocks production approval.

ID	Invariant (non-negotiable)	Primary requirements
R1	Every posted journal balances, is in whole minor units, lies in an open period, is immutable, and its sources reconcile to the GL.	FIN-01..05, INV-02, NFR-DAT-01..03
R2	No one approves, certifies, posts or releases their own work (maker/checker), in the application and in the database.	IAM-08, IAM-09, WFL-01..03, TRS-01
R3	Only authenticated humans decide. Automation, service and AI principals cannot approve, post, pay or release.	IAM-10, WFL-06, AI-02
R4	Tenant, entity and project data are isolated; exports show no more than the screen.	IAM-01, IAM-06, RPT-02, NFR-SEC-06
R5	Exactly one effect per command or event, whatever the retries, duplicates or reordering.	FIN-02, INT-01, INT-02, SIT-02
R6	Offline devices only prepare commands; conflicts are shown, never merged silently.	SIT-02, SIT-04
R7	The assistant sees only what the user may see, uses only computed numbers, and cannot act.	AI-02, AI-03, AI-06, AI-07
R8	No commitment beyond the approved budget without an attested, MFA-backed override.	PCC-01, PCC-02, PRC-03
R9	Tampering is detectable and a backup restores to a consistent, verifiable system.	AUD-01..03, NFR-AVL-02, NFR-AVL-05

3.1 Authoritative write path

The API authenticates the principal and decodes the request. The kernel checks permission, project scope, segregation of duties and business rules, obtains an attestation from the isolated signer for human decisions, and executes one tenant transaction. PostgreSQL then re-verifies the attestation, state machine, balances, periods, budget and tenant identity, and commits the document, its journal (if any), the hash-chained audit record and the outbox event **together or not at all**. Notifications, webhooks, analytics and AI happen only after commit.

3.2 Controlled document lifecycle (reference)

From	To	Trigger / guard
Draft	Submitted	Maker submits; content frozen and attested
Submitted	Approved / Rejected	Checker other than maker; scope, limits, MFA if required; mandatory workflow quorum if published
Approved	Posted / Released / Certified	Further distinct person where configured; period open; budget and policy re-checked
Posted	Reversed	Reversal document by authorised person; original never changes
Any open state	Cancelled	Authorised cancel with reason; commitments released
Offline draft	Recorded / Conflict / Refused	Server revalidation on sync; conflict returned to the author

4. External Interface Requirements

4.1 User interfaces

- **EI-UI-01:** The system shall provide role workspaces (My work, approvals, dashboards) and task-specific screens for every module in Section 6.
- **EI-UI-02:** Each document screen shall show status, history, attachments, related documents and the reason for any refusal.
- **EI-UI-03:** The mobile application shall show clearly whether each record is synced, pending or in conflict.
- **EI-UI-04:** All screens shall meet NFR-USB-01 to NFR-USB-07.

4.2 Hardware interfaces

- **EI-HW-01:** Phone camera and GPS for photos, geotagging and QR or barcode scanning.
- **EI-HW-02:** Biometric attendance terminals through vendor adapters (HCM-03).
- **EI-HW-03:** Equipment telematics through ISO 15143-3 feeds (EQP-07).

4.3 Software and API interfaces

Interface	Examples	Required behaviour
Public REST API	/v1/... (OpenAPI 3)	Authenticated, scoped, rate-limited; idempotency keys on commands
Webhooks	Outbox relay	HMAC-signed, timestamped, ordered per aggregate, replayable
Identity	OIDC, SAML 2.0, SCIM 2.0	Principal kind and MFA assertions; session revocation
Banks	ISO 20022 pain.001 / camt.053, MT940, CSV	Files attested on import; settlement only from statements
Scheduling	Primavera P6 XER/XML, MS Project XML	Import and export with integrity report
BIM	IFC, BCF	View and issue exchange; take-off links
BI	Power BI and warehouse export	Read-only; tenant and project isolation preserved
Offline sync	/v1/sync	Registered device and user token; per-command result
Assistant	/v1/assistant/ask	Facts-only gateway; no write tools

4.4 Communications interfaces

- **EI-CM-01:** HTTPS (TLS 1.2+) only for every external interface.
- **EI-CM-02:** Email (SMTP with DKIM), SMS and messaging gateways through adapters with retry.
- **EI-CM-03:** Controlled outbound egress with timeouts and circuit breakers.

5. Principal Use Cases

UC	Use case	Actor	Main flow (summary)	Related
UC-01	Win and set up a project	BD manager, Estimator, Commercial manager	Opportunity > go/no-go > estimate > tender > award > BOQ approved > baseline budget, WBS and schedule	EST, ORG, PCC, SCH
UC-02	Buy materials for site	Site engineer, Procurement, PM	Requisition > budget check > RFQ > quotes > evaluation > award > PO approval	PRC, PCC
UC-03	Receive, store and issue	Storekeeper, Site engineer	GRN > confirm > stock receipt > issue to cost code > count	PRC, INV, FIN
UC-04	Pay a supplier	AP clerk, AP supervisor, Treasury, CFO	Invoice > three-way match > approval > payment prepare > dual approve > release > bank statement > reconcile	PRC, FIN, TRS
UC-05	Pay a subcontractor	QS, Commercial manager, Treasury	Application > measurement > certificate > retention/advance/back-charges > payable > payment	SUB, FIN, TRS
UC-06	Bill the client and get paid	QS, Commercial manager, AR	Measurement > IPC > client certification > invoice > receipt > retention release	CON, FIN
UC-07	Manage change	PM, QS, Commercial manager	Change event > estimate > client variation and subcontract change > budget revision > schedule update	PCC, CON, SUB, SCH
UC-08	Run the monthly project review	PM, Project controls, Finance	Progress > EVM > forecast to complete > WIP > cost report > approved forecast	EVM, PCC, FIN
UC-09	Record the day on site, offline	Site engineer, Foreman	Diary, timesheets, equipment hours, photos > sync > supervisor approval	SIT, HCM, EQP
UC-10	Close the period	Accountant, Controller, CFO	Subledger reconciliations > accruals > depreciation > payroll > WIP > review TB > close > consolidate	FIN
UC-11	Run payroll	HR officer, Payroll checker, Finance	Timesheets > payroll run > checks > approval > journal > payment file > statutory returns	HCM, FIN, TRS
UC-12	Handle a safety incident	Any worker, HSE officer, PM	Report (offline) > notify > investigate > actions > close > KPIs	HSE
UC-13	Answer an RFI	Site engineer, Consultant	RFI > ball-in-court > response > linked drawing revision > change event if needed	DOC, PCC
UC-14	Ask the assistant	Executive, PM	Question > permission-filtered facts > deterministic answer > sources and assumptions	AI

6. Functional Requirements

Priority: **Must** = required for the release that contains the module; **Should** = important, may be deferred by agreement; **Could** = desirable. Verification: Test, Inspection, Demo, Analysis. Baseline: Exists, Partial, New (see Document Control).

6.1 Identity, Tenancy and Access Control

Construction companies (tenants), their users, roles, scopes, sessions and delegations. Invariants R2, R3, R4.

ID	Requirement	Pri.	Verif.	Baseline
IAM-01	The system shall support multiple tenants (construction companies). Each tenant's data shall be isolated in application code and by PostgreSQL row-level security, with the tenant derived from the database login and never from a client-settable value.	Must	Test	Exists
IAM-02	The system shall authenticate users through enterprise single sign-on using OpenID Connect and SAML 2.0 (for example Microsoft Entra ID, Google Workspace, Okta, Keycloak). Tenants without an identity provider shall be offered local credentials stored only as Argon2id hashes.	Must	Test	Partial
IAM-03	The system shall support SCIM 2.0 user provisioning and deprovisioning; a deprovisioned user's sessions shall be revoked within 60 seconds.	Should	Test	New
IAM-04	The system shall require multi-factor authentication (WebAuthn/passkeys or TOTP; SMS only if the tenant enables it) for every user holding an approval, posting, payment, security or administration permission, and step-up MFA bound to the specific action within a configurable freshness window (default 10 minutes).	Must	Test	Partial
IAM-05	The system shall provide role-based access control with the seed role catalogue of Appendix D, and shall let tenants compose custom roles from the fixed permission list. A role that would hold both the maker and the checker permission of the same control shall be refused.	Must	Test	Exists
IAM-06	The system shall restrict every read and command to the projects explicitly assigned to the user, unless the user holds a tenant-wide scope permission.	Must	Test	Exists
IAM-07	The system shall evaluate attribute-based rules (legal entity, business unit, project, amount band, document classification, device, principal kind) on every command and read in addition to role checks.	Must	Test	Partial
IAM-08	The system shall enforce segregation of duties (maker differs from checker, and checker differs from poster or releaser where configured) for every controlled document, both in the application and by database constraint.	Must	Test	Exists
IAM-09	The system shall analyse segregation-of-duties conflicts before a role is assigned, against a tenant-maintained conflict rule set, and shall require a documented mitigating control to accept a conflict.	Should	Test	Partial
IAM-10	The system shall distinguish human, service and AI principals from an identity-provider assertion that no role can grant. Only human principals shall approve, certify, post, pay, release, close or configure.	Must	Test	Exists
IAM-11	The system shall revoke sessions server-side across all API replicas; a role removal shall take effect on the next request.	Must	Test	Exists
IAM-12	The system shall rate-limit authentication and API use with limits shared across replicas, and shall temporarily lock an account after a configurable number of failures and notify the tenant security administrator.	Must	Test	Partial
IAM-13	The system shall maintain a delegation-of-authority matrix (approval limits by role, document type, amount band, entity and project). The matrix shall be versioned and approved by maker/checker.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
IAM-14	The system shall support time-bound, scoped delegation of approval rights (for example during leave). A delegate shall never exceed the delegator's limits, and delegation shall never enable self-approval.	Should	Test	Exists
IAM-15	The system should provide time-limited break-glass access that requires a typed reason, notifies security administrators and is flagged for review.	Should	Demo	New
IAM-16	The system may support tenant conditional-access policies (IP allow-lists, device posture, country restrictions).	Could	Test	New

6.2 Organisation and Master Data

Group structure, projects, business partners, cost structures and resources. Atlas references: modules Ledger, WBS, P2P.

ID	Requirement	Pri.	Verif.	Baseline
ORG-01	The system shall model group, legal entity, region or branch, business unit and project. Each legal entity shall have its own functional currency, tax registration, chart-of-accounts mapping and fiscal calendar.	Must	Test	Partial
ORG-02	The system shall maintain a project master: code, name, client, contract type, location (with GPS coordinates), key dates, project manager, phase, status and stage-gate history.	Must	Test	Exists
ORG-03	The system shall maintain supplier and subcontractor masters with maker/checker approval and blocking. Bank accounts shall be dual-approved with MFA and shown masked unless the user holds the reveal permission (which is audited).	Must	Test	Exists
ORG-04	The system shall maintain a customer master with TIN, credit limit, payment terms and maker/checker approval.	Must	Test	Partial
ORG-05	The system should support supplier and subcontractor pre-qualification: trade licence and contractor grade, tax clearance, VAT registration, insurance and expiry tracking, with alerts before expiry and an evaluated performance rating.	Should	Test	New
ORG-06	The system shall maintain versioned WBS and cost-code structures with maker/checker approval. Documents shall only reference cost codes in an approved version.	Must	Test	Exists
ORG-07	The system shall maintain a resource library (labour trades, plant types, materials, subcontract packages) with units of measure and conversion factors.	Must	Test	Partial
ORG-08	The system should detect probable duplicate partners (same TIN, bank account, or similar name) at creation and route them to review.	Should	Test	New
ORG-09	The system shall maintain working calendars per entity and project, including Ethiopian public holidays, shifts and non-working days.	Must	Test	Partial

6.3 Business Development, Estimating and Tendering

From opportunity to an approved bid and baseline budget.

ID	Requirement	Pri.	Verif.	Baseline
EST-01	The system should manage an opportunity and bid pipeline: source, client, value, go/no-go scoring, probability, expected value and bid calendar.	Should	Test	New
EST-02	The system shall manage bills of quantities (BOQ) in versions with items, units, quantities and rates mapped to cost codes. A submitted version shall be immutable; approval shall be by a different person.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
EST-03	The system shall support resource-based estimating: unit-rate build-ups from labour, plant, materials and subcontract resources with productivity factors, plus indirect costs, overhead, risk and margin, drawing on a maintained cost database.	Must	Test	New
EST-04	The system should compare estimate versions and markup scenarios, and price imported resources in their source currency using approved FX rates.	Should	Test	New
EST-05	The system should support quantity take-off from 2D drawings and IFC models with each quantity linked to a BOQ item.	Should	Demo	New
EST-06	The system should manage tender packages: documents, clarifications, addenda, bid-bond tracking and a submission checklist.	Should	Test	New
EST-07	An approved BOQ shall become the project's baseline budget through a controlled transition. It shall never be changed in place.	Must	Test	Exists
EST-08	The system may benchmark unit rates and productivity against completed projects.	Could	Demo	New

6.4 Project Controls: Budget, Commitment, Cost and Forecast

The core of a construction ERP: knowing at any moment what each project was budgeted, has committed, has spent and will cost. Invariant R8.

ID	Requirement	Pri.	Verif.	Baseline
PCC-01	The system shall keep the original budget, approved budget revisions, budget transfers between cost codes, and contingency and management-reserve draw-downs, each approved by maker/checker. Original and current budgets shall always be shown side by side.	Must	Test	Exists
PCC-02	The system shall record commitments from approved purchase orders and subcontracts in real time. It shall refuse a requisition or order above the remaining budget of its cost code unless an authorised person overrides it with fresh MFA and a reason.	Must	Test	Exists
PCC-03	The system shall derive actual cost from posted journals by project and cost code. Each figure shall drill down to its source documents.	Must	Test	Exists
PCC-04	The system shall support periodic cost forecasting: estimate to complete and estimate at completion per cost code (manual, CPI-based or remaining-budget methods), forecast versions and an approved monthly forecast cycle.	Must	Test	Partial
PCC-05	The system shall produce a cost report showing, per cost code: original budget, revised budget, committed, actual, accrued, forecast, and variances against budget and the last forecast.	Must	Test	Partial
PCC-06	The system shall produce a work-in-progress schedule per project (percentage complete, earned revenue, billed to date, over- or under-billing) that feeds revenue recognition (FIN-15).	Must	Test	New
PCC-07	The system should manage change events (potential, pending, approved) with cost and schedule impact, linked to client variations and subcontract changes.	Should	Test	New
PCC-08	The system should keep a quantified risk and opportunity register per project and compute contingency at P50 and P80 by Monte Carlo simulation.	Should	Test	New
PCC-09	The system shall roll up project figures by entity, region, business unit, client, contract type and portfolio.	Must	Test	Partial
PCC-10	The system should show project cash flow as planned, actual and forecast S-curves.	Should	Test	New

6.5 Planning and Scheduling

Critical-path scheduling integrated with cost and progress. Atlas reference: ADR 0013.

ID	Requirement	Pri.	Verif.	Baseline
SCH-01	The system shall compute critical-path schedules with FS, SS, FF and SF relationships, lags, activity calendars and total and free float.	Must	Test	Exists
SCH-02	The system shall keep approved baselines (maker/checker) and compare the current schedule against any of several baselines.	Must	Test	Partial
SCH-03	The system shall support schedule constraints (start no earlier than, finish no later than, mandatory), activity codes and a WBS hierarchy.	Must	Test	New
SCH-04	The system should load resources onto activities, level them, and show labour and plant histograms.	Should	Test	New
SCH-05	The system shall import and export Primavera P6 (XER and XML) and Microsoft Project XML, with an integrity report of anything not carried over.	Must	Test	New
SCH-06	The system should support look-ahead planning (3 to 6 weeks) and weekly work plans with percent-plan-complete tracking.	Should	Test	New
SCH-07	The system should support delay analysis: time-impact fragnets and as-planned versus as-built comparison, linked to extension-of-time claims (CON-04).	Should	Test	New
SCH-08	The system should link schedule activities to model elements for 4D visualisation.	Should	Demo	New
SCH-09	The system may run a schedule quality check (for example the DCMA 14-point assessment).	Could	Test	New

6.6 Progress Measurement and Earned Value

Objective measurement of performance. Atlas reference: ADR 0013.

ID	Requirement	Pri.	Verif.	Baseline
EVM-01	The system shall record progress reports per activity or cost code, approved by a person other than the author.	Must	Test	Exists
EVM-02	The system shall compute PV, EV, AC, SV, CV, SPI, CPI, EAC, ETC and VAC with a documented price basis (nominal or constant) and provenance for every input.	Must	Test	Exists
EVM-03	The system should support rules of credit: weighted milestones, units installed, percent complete, start/finish and level of effort, configurable per cost code.	Should	Test	New
EVM-04	The system should feed installed quantities captured on site (SIT) into progress and client measurement.	Should	Test	New
EVM-05	The system should attach photo and geotag evidence to progress records.	Should	Test	New

6.7 Procurement and Supply Chain

Procure-to-pay from requisition to payment. Atlas reference: Journey A, ADR 0009. Invariants R2, R8.

ID	Requirement	Pri.	Verif.	Baseline
PRC-01	The system shall support the flow requisition, RFQ, quotes, technical evaluation, award, purchase order, goods receipt and three-way match, with each step a controlled document.	Must	Test	Exists
PRC-02	The system shall enforce a minimum number of suppliers per RFQ unless there is a recorded single-source justification. An award to other than the lowest compliant quote shall require a reason.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
PRC-03	Purchase-order approval shall follow the delegation-of-authority matrix and budget control (PCC-02).	Must	Test	Exists
PRC-04	The system should support framework agreements and blanket orders with call-offs against agreed price lists.	Should	Test	New
PRC-05	The system should provide e-sourcing through the supplier portal. Quotes shall be sealed until the due time and opened by two people.	Should	Test	New
PRC-06	The system shall manage import procurement: proforma, letter of credit (TRS-05), shipping documents, customs clearance, and allocation of landed cost (freight, insurance, duty, clearing) to received items.	Must	Test	New
PRC-07	The system should track expediting and delivery, and score supplier performance on price, time and quality.	Should	Test	New
PRC-08	The system shall flag duplicate supplier invoices and split purchases just below approval thresholds.	Must	Test	Exists
PRC-09	The system should version purchase-order changes, with re-approval when value, quantity or terms change.	Should	Test	New

6.8 Subcontract Management

Subcontractors typically account for most of a contractor's cost; they need their own controlled lifecycle.

ID	Requirement	Pri.	Verif.	Baseline
SUB-01	The system shall manage subcontract agreements with a schedule of values or rates, retention percentage, advance payment, scope and terms. They shall commit against the budget (PCC-02).	Must	Test	New
SUB-02	The system shall process subcontractor payment applications: measured work, certification by a quantity surveyor other than the preparer, retention, advance recovery, back-charges, withholding tax, and generation of the payable.	Must	Test	New
SUB-03	The system shall manage subcontract variations linked to change events (PCC-07).	Must	Test	New
SUB-04	The system should track subcontractor compliance (insurance, licences, safety induction) and, if the tenant configures it, block payment while a document is expired.	Should	Test	New
SUB-05	The system should record back-charges and contra-charges with evidence and the subcontractor's acknowledgement.	Should	Test	New
SUB-06	The system should support pay-when-certified rules that link subcontractor payment to client certification.	Should	Test	New
SUB-07	The system should let subcontractors submit applications and documents through the subcontractor portal (COL-03).	Should	Demo	New

6.9 Inventory and Materials

Central warehouses and site stores. Atlas reference: Journey C, ADR 0012.

ID	Requirement	Pri.	Verif.	Baseline
INV-01	The system shall manage warehouses and site stores, with receipts, issues to project cost codes, and approved transfers between stores.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
INV-02	The system shall value stock by FIFO or weighted average per tenant policy and reconcile the stock subledger to the inventory control account with zero unexplained difference.	Must	Test	Exists
INV-03	The system shall support stock counts with approval. Adjustments shall generate journals approved and posted by people other than the counter.	Must	Test	Exists
INV-04	The system shall convert units of measure (for example bag to tonne, piece to metre) and store stock in a base unit.	Must	Test	Partial
INV-05	The system should support material requests and reservations against schedule activities.	Should	Test	Exists
INV-06	The system should track batches, heat numbers and serial numbers (for example a rebar heat or cement batch) with linked test certificates (QA-04).	Should	Test	New
INV-07	The system should support reorder points, minimum and maximum levels per store, and suggested requisitions.	Should	Test	Exists
INV-08	The system should support returns to supplier and transfers of surplus between projects at valuation.	Should	Test	New
INV-09	The system should compare material consumption with the theoretical quantity from the BOQ and report wastage.	Should	Test	New
INV-10	The system may print barcode or QR labels and support mobile scanning.	Could	Demo	New

6.10 Equipment and Fleet

Owned, leased and hired plant as a cost and revenue centre.

ID	Requirement	Pri.	Verif.	Baseline
EQP-01	The system shall keep an equipment register (owned, leased, hired) linked to fixed assets (FIN-06), with ownership and operating cost.	Must	Test	New
EQP-02	The system shall charge internal hire rates to projects by recorded usage (hours, days or kilometres) through generated journals.	Must	Test	New
EQP-03	The system shall record daily working, idle and breakdown hours per item of equipment, captured in site diaries or timesheets.	Must	Test	Partial
EQP-04	The system should plan preventive maintenance (by hours, distance or calendar), raise work orders and issue spare parts from inventory.	Should	Test	New
EQP-05	The system should record fuel issues per item of equipment and flag consumption anomalies.	Should	Test	New
EQP-06	The system should record equipment transfers between sites with a condition report at each hand-over.	Should	Test	New
EQP-07	The system may import telematics data (ISO 15143-3 / AEMP 2.0).	Could	Demo	New

6.11 Workforce, Time and Payroll

Salaried staff and daily labour, costed to projects.

ID	Requirement	Pri.	Verif.	Baseline
HCM-01	The system shall keep an employee master covering salaried staff and daily labourers: contracts, grade, trade, rate, certifications and bank or mobile-money details (masked).	Must	Test	New

ID	Requirement	Pri.	Verif.	Baseline
HCM-02	The system shall capture timesheets per person or crew against project and cost code, including offline, approved by a supervisor other than the person recording them.	Must	Test	New
HCM-03	The system should capture attendance from biometric devices or a geofenced mobile check-in.	Should	Test	New
HCM-04	The system shall compute payroll with data-driven, effective-dated statutory rules (Ethiopian employment income tax bands, employee and employer pension contributions, overtime under the labour law in force) confirmed by a qualified adviser.	Must	Test	New
HCM-05	Payroll shall be approved by maker/checker and shall generate journals by project and cost code, a bank or mobile-money payment file, and statutory returns.	Must	Test	New
HCM-06	The system should manage leave and overtime requests and approvals.	Should	Test	New
HCM-07	The system should report labour productivity (units installed per person-hour) against the estimate.	Should	Test	New
HCM-08	The system may support recruitment and onboarding of site workers, with induction records linked to HSE.	Could	Demo	New

6.12 Contracts, Revenue and Claims

Client side: contract to cash. Atlas reference: Journey B, ADR 0011.

ID	Requirement	Pri.	Verif.	Baseline
CON-01	The system shall manage client contracts with lines or rates, payment terms and a contract policy (retention, advance, VAT base), approved by maker/checker.	Must	Test	Exists
CON-02	The system shall support measurement, interim payment certificate (IPC), receivable invoice and receipt, with retention, retention release and advance recovery computed by the server.	Must	Test	Exists
CON-03	The system shall manage client variations with approval and their effect on contract value.	Must	Test	Exists
CON-04	The system shall keep a claims and extension-of-time register with contractual notices, time bars (for example a 28-day notice period) and alerts before a time bar expires.	Must	Test	New
CON-05	The system shall apply contract price-adjustment formulas based on published indices, per contract.	Must	Test	New
CON-06	The system should compute liquidated damages and bonuses from contract dates.	Should	Test	New
CON-07	The system shall keep a register of guarantees and bonds (bid, advance-payment, performance, retention) with issuer, amount, expiry, reduction and release, and alert before expiry.	Must	Test	New
CON-08	The system should provide configurable contract templates (FIDIC Red, Yellow and Silver Books; Ethiopian public-procurement standard documents) that set default policies.	Should	Test	New
CON-09	The system should keep a contract obligations register with owners and due dates.	Should	Test	New
CON-10	The system should let clients and consultants review measurements and IPCs through the client portal (COL-03).	Should	Demo	New

6.13 Finance and Accounting

The deterministic financial core. Atlas references: ADRs 0002, 0003, 0009. Invariant R1.

ID	Requirement	Pri.	Verif.	Baseline
FIN-01	The system shall record journals as balanced, immutable double entries in whole minor units, with preparer, approver and poster distinct. Corrections shall be by reversal only.	Must	Test	Exists
FIN-02	Each journal generated from a business document shall carry a unique source-event identifier so that no event can be posted twice.	Must	Test	Exists
FIN-03	The system shall manage accounting periods. Posting into a closed period shall be refused, and reopening shall need a privileged, audited permission.	Must	Test	Exists
FIN-04	The system shall support multi-currency transactions with approved rates selected by an approved FX policy, and remeasurement under IAS 21.	Must	Test	Exists
FIN-05	The system shall provide an accounts payable subledger and an accounts receivable subledger with aging, reconciled to their control accounts.	Must	Test	Exists
FIN-06	The system shall manage fixed assets under IAS 16: acquisition, components, depreciation methods, transfers, revaluation, impairment and disposal.	Must	Test	New
FIN-07	The system shall apply data-driven, effective-dated tax rules: output and input VAT, withholding tax on payments and receipts, and excise where applicable. It shall produce the supporting schedules for tax returns.	Must	Test	Partial
FIN-08	The system shall consolidate multiple entities, with intercompany transactions matched and eliminated automatically, and joint ventures accounted for under the tenant's policy.	Must	Test	New
FIN-09	The system shall produce trial balance, general ledger, income statement, balance sheet and cash-flow statement per entity and consolidated, in IFRS presentation.	Must	Test	Partial
FIN-10	The system should support recurring journals, and accruals and prepayments with automatic reversal.	Should	Test	New
FIN-11	The system should allocate overheads to projects by configurable rules.	Should	Test	New
FIN-12	The system should support corporate (overhead) budgets with variance reporting.	Should	Test	New
FIN-13	The system should support the Ethiopian fiscal year (Hamle to Sene) and display dates in the Ethiopian calendar.	Should	Test	New
FIN-14	The system should provide an adapter for tax-authority e-invoicing and fiscal-receipt obligations, once their applicability and format are confirmed (Appendix E).	Should	Analysis	New
FIN-15	The system shall recognise contract revenue over time under IFRS 15 (input or output method per contract) and maintain contract assets and liabilities.	Must	Test	New

6.14 Treasury and Cash Management

Payments as records, settlement from the bank. Atlas reference: Journey A steps 15-19.

ID	Requirement	Pri.	Verif.	Baseline
TRS-01	Payments shall be prepared, approved (dual approval above a threshold) and released by different human principals with fresh MFA. The system shall never move money itself; settlement shall be recognised only from imported bank statements.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
TRS-02	The system shall forecast cash for 13 weeks and for the project horizon, separating confirmed from forecast flows and including payroll, taxes, supplier schedules, expected IPC receipts and debt service.	Must	Test	Partial
TRS-03	The system should exchange payment files (ISO 20022 pain.001) and statements (camt.053, MT940 or bank CSV) with banks where they support it.	Should	Test	New
TRS-04	The system should track bank facilities, overdrafts and guarantee lines, and their utilisation.	Should	Test	New
TRS-05	The system should manage letters of credit for imports, from application to settlement.	Should	Test	New
TRS-06	The system may manage site petty cash on an imprest basis, with replenishment approvals.	Could	Test	New

6.15 Site Operations and Mobile

Work where it happens, on intermittent connectivity. Atlas reference: ADR 0017. Invariant R6.

ID	Requirement	Pri.	Verif.	Baseline
SIT-01	The system shall record daily site diaries (weather, labour, equipment, notes, instructions), approved by a person other than the author.	Must	Test	Exists
SIT-02	Only registered, revocable devices shall sync offline work. Sync shall be idempotent per device and operation id, and a conflict shall be returned to the user, never merged silently.	Must	Test	Exists
SIT-03	The system shall provide an offline-first mobile application (Android first, then iOS) for diaries, timesheets, goods receipts, inspections, incidents, photos and approvals.	Must	Demo	New
SIT-04	Offline devices shall prepare commands only. They shall never create authoritative financial state.	Must	Test	Exists
SIT-05	The system should capture photos with GPS position, time and watermark, and link them to any record.	Should	Test	New
SIT-06	The system should manage punch and snag lists with assignees, due dates and close-out photos.	Should	Test	New
SIT-07	The system should record site instructions and meeting minutes with tracked actions.	Should	Test	New
SIT-08	The system may link drone or 360-degree captures to locations and dates.	Could	Demo	New

6.16 Document Control and BIM

A common data environment for drawings, models and correspondence.

ID	Requirement	Pri.	Verif.	Baseline
DOC-01	The system shall provide a common data environment following ISO 19650 (work in progress, shared, published, archived), with tenant-defined numbering rules and metadata.	Must	Test	New
DOC-02	The system shall manage document revisions. A superseded revision shall never be presented as current.	Must	Test	New
DOC-03	The system shall issue transmittals and record each recipient's acknowledgement.	Must	Test	New

ID	Requirement	Pri.	Verif.	Baseline
DOC-04	The system shall manage RFIs, with ball-in-court, due dates, responses, and links to cost and schedule impact.	Must	Test	New
DOC-05	The system should manage submittals (shop drawings, material approvals) with review status and stamps.	Should	Test	New
DOC-06	The system should keep a drawing register and allow mark-up in the browser.	Should	Demo	New
DOC-07	The system should view IFC models in the browser and exchange model-based issues in BCF.	Should	Demo	New
DOC-08	Files shall be encrypted at rest, scanned for malware and served only through short-lived signed URLs after an authorisation check.	Must	Test	New
DOC-09	The system should provide permission-filtered full-text search over documents, including text extracted by OCR.	Should	Test	New

6.17 Quality Management

Inspection and test plans, non-conformances, material tests.

ID	Requirement	Pri.	Verif.	Baseline
QA-01	The system shall manage inspection and test plans per activity or work package, with hold and witness points.	Must	Test	New
QA-02	The system shall record inspection requests and results. If the tenant configures it, progress or measurement on an activity with an unreleased hold point shall be refused.	Must	Test	New
QA-03	The system shall manage non-conformance reports with root cause, corrective and preventive action, cost impact and verified closure.	Must	Test	New
QA-04	The system should record material test results (for example concrete cubes or rebar tensile tests) linked to batches (INV-06).	Should	Test	New
QA-05	The system should feed quality outcomes into supplier and subcontractor ratings.	Should	Test	New

6.18 Health, Safety and Environment

Safety records that are timely and impossible to quietly lose.

ID	Requirement	Pri.	Verif.	Baseline
HSE-01	The system shall record incidents and near misses (including offline), with investigation, root cause and corrective actions tracked to closure.	Must	Test	New
HSE-02	The system shall manage permits to work (hot work, confined space, work at height, excavation, lifting) with approvals, conditions and expiry.	Must	Test	New
HSE-03	The system should record inductions, toolbox talks and training, and can be configured to block site access or timesheets for workers who have not been inducted.	Should	Test	New
HSE-04	The system should run safety inspection and observation checklists.	Should	Test	New
HSE-05	The system shall compute HSE indicators (LTIFR, TRIR, severity rate) using hours worked from timesheets.	Must	Test	New
HSE-06	The system may record environmental aspects: waste, spills and environmental permits.	Could	Test	New

6.19 Workflow, Approvals and Automation

Configurable routing in which only people decide. Atlas references: ADRs 0007, 0008, 0015. Invariants R2, R3.

ID	Requirement	Pri.	Verif.	Baseline
WFL-01	The system shall execute versioned workflow definitions declaring steps, permissions, quorums, SoD rules, MFA, amount thresholds, SLA escalation and expiry. A validator shall refuse any definition that weakens a control.	Must	Test	Exists
WFL-02	When a definition is published as mandatory, the database shall refuse the subject's sensitive transition unless the same transaction recorded the workflow effect with its full quorum.	Must	Test	Exists
WFL-03	Every human decision shall carry an attestation signed by the isolated approval signer and verified by the database.	Must	Test	Exists
WFL-04	The system shall present one task inbox across all modules, on web and mobile, with step-up MFA for sensitive decisions.	Must	Test	Partial
WFL-05	The system should provide a visual workflow designer that produces the same validated definition format.	Should	Test	New
WFL-06	Automation rules shall only notify or prepare drafts. The rule's author and its activator shall be different people.	Must	Test	Exists
WFL-07	The system should redirect tasks on absence through approved delegations (IAM-14).	Should	Test	New

6.20 Reporting, Dashboards and Analytics

Trusted numbers, traced to their documents.

ID	Requirement	Pri.	Verif.	Baseline
RPT-01	The system shall provide a maintained catalogue of standard reports across all modules (at least 60 at general availability; currently 11 of the 25 initially listed exist).	Must	Test	Partial
RPT-02	Every report and list shall export to PDF, Word and Excel with the tenant's letterhead. Exports shall contain no more than the user can see, and shall be watermarked and audited.	Must	Test	Exists
RPT-03	The system shall provide role dashboards (executive, project, finance, procurement, HSE). Every KPI shall have a documented formula and drill down to its source documents.	Must	Test	Partial
RPT-04	The system should offer a self-service report builder over a governed semantic layer, with saved filters and views.	Should	Test	New
RPT-05	The system should deliver reports by email on a schedule, subject to the recipient's permissions.	Should	Test	New
RPT-06	Every report shall display its filters and data-freshness timestamp.	Must	Test	Partial
RPT-07	The system should provide a read-only analytics feed (warehouse export and BI connectors such as Power BI) that preserves tenant and project isolation.	Should	Test	New
RPT-08	The system should produce statutory reports: VAT return, withholding tax, pension and payroll tax schedules.	Should	Test	New

6.21 Embedded Intelligence and AI

AI that advises and drafts but never decides. Atlas reference: platform 0038. Invariant R7.

ID	Requirement	Pri.	Verif.	Baseline
AI-01	The system should answer operational questions (cost overrun, schedule slip, cash, overdue orders) from permission-filtered facts, citing sources, freshness, assumptions and missing data.	Should	Demo	Partial
AI-02	AI principals shall not approve, certify, post, pay or release. Their only write capability shall be preparing drafts for a human.	Must	Test	Exists
AI-03	All numbers shown shall come from deterministic computation. Any language-model phrasing shall be checked against the facts and discarded if it introduces a number not in them.	Must	Test	Exists
AI-04	The system should extract data from supplier invoices, delivery notes and certificates into draft documents with per-field confidence. A person shall confirm each draft.	Should	Test	New
AI-05	The system should predict cost-overrun, schedule-slip, cash-shortfall and supplier-delay risk with explanations, and record each prediction so it can be compared with the rule-based baseline and with outcomes.	Should	Test	Partial
AI-06	Untrusted content (documents, OCR output, free text) shall be treated as data, never as instructions. Prompt-injection red-team tests shall run in CI.	Must	Test	Partial
AI-07	Use of an external model shall require tenant opt-in, data minimisation and an approved data-residency decision. A tenant administrator shall be able to disable all AI features.	Must	Inspection	New
AI-08	AI interactions (user, question, facts used, timestamp) shall be logged without unnecessary personal data.	Must	Inspection	Partial

6.22 Audit, Compliance and Traceability

Evidence that survives an auditor and an attacker. Invariant R9.

ID	Requirement	Pri.	Verif.	Baseline
AUD-01	The system shall write a hash-chained, append-only audit record for every command, decision, export, sensitive-data reveal, login and configuration change.	Must	Test	Exists
AUD-02	Every business change shall write its audit record and outbox event in the same transaction.	Must	Test	Exists
AUD-03	Posted records shall be immutable to every application login and to the schema owner (database triggers). An integrity scan shall detect out-of-band tampering.	Must	Test	Exists
AUD-04	Audit logs shall be readable only by authorised roles, and reading them shall itself be audited. External auditors shall get time-boxed, read-only access.	Must	Test	Partial
AUD-05	The system shall provide a control review report: SoD exceptions, budget overrides, approvals outside SLA, reopened periods and revealed bank details.	Must	Test	Partial
AUD-06	The system shall trace end to end from payment to invoice, receipt, purchase order, award, requisition, budget line and BOQ item, and from receipt to IPC, measurement and contract line.	Must	Test	Partial
AUD-07	The system should generate a period evidence pack for the external auditor.	Should	Analysis	New

6.23 Collaboration, Notifications and Portals

Bringing clients, consultants, suppliers and subcontractors into controlled workflows.

ID	Requirement	Pri.	Verif.	Baseline
COL-01	The system shall raise in-app notifications from outbox events after commit.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
COL-02	The system should deliver notifications by email, SMS and messaging channels (for example Telegram), configurable by role and user preference, with retry.	Should	Test	New
COL-03	The system should provide external portals for clients, consultants, suppliers and subcontractors. Portal identities shall be limited to their own records and actions.	Should	Test	New
COL-04	The system should support comments and mentions on any record, retained with the record's audit trail.	Should	Test	New

6.24 Integration and Interoperability

Open by design, so that no tenant is locked in.

ID	Requirement	Pri.	Verif.	Baseline
INT-01	All APIs shall be versioned, documented in OpenAPI 3, authenticated and rate-limited. Every command shall accept an idempotency key.	Must	Inspection	Exists
INT-02	Domain events shall be delivered as signed webhooks from the transactional outbox, with per-aggregate ordering, retries, a dead-letter queue and replay.	Must	Test	Exists
INT-03	The system should offer bulk import and export (CSV and Excel templates) through a validated staging area with review before commit.	Should	Test	Partial
INT-04	The system should provide connectors for Primavera P6, Microsoft Project, Microsoft 365 and SharePoint, Google Drive and Power BI.	Should	Test	New
INT-05	The system should connect to Ethiopian commercial banks and to mobile-money services (for example telebirr) where they offer interfaces.	Should	Test	New
INT-06	External adapters shall be isolated behind timeouts, retries with back-off and circuit breakers, so that no outside failure blocks internal transactions.	Must	Test	Partial
INT-07	The system should provide migration tooling from common legacy systems (Peachtree/Sage, QuickBooks, Excel, SAP Business One), with reconciliation reports.	Should	Test	New

6.25 Administration, Configuration and Localisation

Tenant self-service without code.

ID	Requirement	Pri.	Verif.	Baseline
ADM-01	The system shall provide guided tenant onboarding: entities, chart of accounts (construction IFRS template), taxes, banks, users, roles and policies.	Must	Demo	Partial
ADM-02	Operational policies (P2P, contract, stock, FX, thresholds) shall be versioned, maker/checker approved and audited.	Must	Test	Exists
ADM-03	The system shall manage company settings: letterhead, document numbering, locale and portal branding.	Must	Test	Partial
ADM-04	The interface shall be available in English and Amharic at launch. Afaan Oromo and Tigrinya shall be addable without code changes. Translation shall never alter canonical data.	Must	Test	Partial
ADM-05	The system should let administrators add custom fields per document type, which shall be validated and available in reports.	Should	Test	New
ADM-06	The system should show administrators backup, queue, sync and integration health.	Should	Demo	Partial

ID	Requirement	Pri.	Verif.	Baseline
ADM-07	The system should create sandbox copies of a tenant, with personal data masked, for training and testing.	Should	Test	New
ADM-08	The system shall support configurable retention periods and legal holds per record type.	Must	Test	New

7. Non-Functional Requirements

7.1 Performance and Capacity

Proposed baselines for agreement; validated by load tests on a production-like environment (phase R4). Server-side, excluding client network latency.

ID	Requirement	Pri.	Verif.	Baseline
NFR-PERF-01	Command endpoints shall respond within 1 s at the 95th percentile and 3 s at the 99th percentile under the reference load.	Must	Test	New
NFR-PERF-02	Lists and detail reads shall respond within 500 ms at the 95th percentile. Search shall respond within 1 s over 5 million documents per tenant.	Must	Test	New
NFR-PERF-03	Reports over up to 100,000 rows shall complete within 10 s. Larger reports shall run asynchronously and notify the user.	Should	Test	New
NFR-PERF-04	The reference deployment shall support 2,000 concurrent users, 500 active projects per tenant and 50 million journal lines per tenant without breaching NFR-PERF-01.	Must	Test	New
NFR-PERF-05	A device with 1,000 queued offline commands shall finish syncing within 5 minutes on a 3G-class connection.	Should	Test	New
NFR-PERF-06	Over a constrained connection, the web console shall load in under 1.5 MB (compressed) on first visit.	Should	Test	New

7.2 Availability, Reliability and Recovery

Invariant R9.

ID	Requirement	Pri.	Verif.	Baseline
NFR-AVL-01	The SaaS service shall achieve 99.9 % monthly availability for transactional services, excluding announced maintenance (99.5 % during the first six months after launch).	Must	Analysis	New
NFR-AVL-02	The recovery point objective shall be at most 5 minutes (point-in-time recovery) and the recovery time objective at most 4 hours, or 1 hour for the premium tier.	Must	Test	Partial
NFR-AVL-03	An interruption at any point shall leave each transaction fully committed or fully rolled back. Replaying the request shall be safe.	Must	Test	Exists
NFR-AVL-04	A failure of reporting, AI, notification, export or integration services shall not prevent transactions.	Must	Test	Partial
NFR-AVL-05	Backups shall be encrypted, copied to an isolated off-site location in object-locked storage, and restore-tested at least quarterly, with a signed record of each test.	Must	Test	Partial
NFR-AVL-06	Deployments and migrations shall be backward-compatible so that releases need no downtime during working hours.	Should	Inspection	New

7.3 Security

Baseline standard: OWASP ASVS. Invariants R2, R3, R4, R7.

ID	Requirement	Pri.	Verif.	Baseline
NFR-SEC-01	All traffic shall use TLS 1.2 or higher (TLS 1.3 preferred), with HSTS on public endpoints.	Must	Inspection	Partial

ID	Requirement	Pri.	Verif.	Baseline
NFR-SEC-02	Data at rest shall be encrypted. Bank details and personal identifiers shall also have field-level encryption.	Must	Inspection	Partial
NFR-SEC-03	The system shall meet OWASP ASVS Level 2, with Level 3 for authentication, approvals, payments and administration.	Must	Analysis	Partial
NFR-SEC-04	The approval-signing key shall be held in an HSM or cloud KMS under a documented key ceremony. Other secrets shall be held in a vault and rotated. No secret shall be in source control or client code.	Must	Inspection	Partial
NFR-SEC-05	The CI pipeline shall run SAST, dependency vulnerability scanning (govulncheck), secret scanning, checksum verification and SBOM generation, and shall sign container images (target SLSA level 3).	Must	Inspection	Partial
NFR-SEC-06	Automated cross-tenant and privilege-escalation probes shall run in CI against the database as a tenant login and as the schema owner.	Must	Test	Exists
NFR-SEC-07	An independent penetration test shall close all critical and high findings before general availability, and shall be repeated at least annually.	Must	Analysis	New
NFR-SEC-08	Security events shall be logged centrally (SIEM export) with alerts. Logs shall never contain passwords, tokens or bank details.	Must	Inspection	Partial
NFR-SEC-09	Exports shall be subject to per-user volume budgets, watermarks and anomaly alerts (anti-exfiltration).	Must	Test	Partial
NFR-SEC-10	The vendor should obtain ISO/IEC 27001 certification and a SOC 2 Type II report within 18 months of general availability.	Should	Analysis	New

7.4 Privacy and Regulatory Compliance

Not legal advice: applicability shall be confirmed by qualified counsel.

ID	Requirement	Pri.	Verif.	Baseline
NFR-PRV-01	The system shall support compliance with the Ethiopian Personal Data Protection Proclamation No. 1321/2024 (minimisation, purpose limitation, data-subject rights, breach notification, cross-border restrictions), and with GDPR for tenants that need it.	Must	Analysis	New
NFR-PRV-02	Tenants shall be able to choose in-country hosting or on-premises deployment (data residency).	Must	Analysis	New
NFR-PRV-03	Retention periods for financial, payroll, contract and HSE records shall be set to legal minimums confirmed by counsel. Deletion shall be by approved purge or anonymisation only.	Must	Analysis	New
NFR-PRV-04	A documented incident and breach-response procedure, and a data-processing agreement template, shall exist before general availability.	Must	Inspection	New
NFR-PRV-05	Financial and tax treatment (IFRS, VAT, withholding, payroll) shall be reviewed and signed by a qualified accountant or tax adviser before live use.	Must	Analysis	Partial

7.5 Usability, Accessibility and Localisation

ID	Requirement	Pri.	Verif.	Baseline
NFR-USB-01	Non-mobile screens shall conform to WCAG 2.1 level AA, and every function shall be usable by keyboard.	Must	Inspection	Partial
NFR-USB-02	A clerk shall be able to work a standard document (requisition, goods receipt, invoice) after no more than one day of training, to be validated in UAT.	Should	Demo	New

ID	Requirement	Pri.	Verif.	Baseline
NFR-USB-03	Layouts shall adapt from desktop to tablet and phone. Approvals and dashboards shall be fully usable on a phone.	Must	Test	Partial
NFR-USB-04	Ethiopic script shall render correctly on screen and in every export format.	Must	Test	Partial
NFR-USB-05	Users should be able to display dates in the Ethiopian or Gregorian calendar. Timestamps shall be stored in UTC and displayed by default in the Africa/Addis_Ababa time zone.	Should	Test	New
NFR-USB-06	The server's refusals shall be explained in plain language that names the rule and the action that would satisfy it.	Must	Test	Exists
NFR-USB-07	The product should apply one documented design system across web, mobile and portals.	Should	Inspection	New

7.6 Maintainability, Portability and Operations

Atlas references: ADRs 0001 and 0016.

ID	Requirement	Pri.	Verif.	Baseline
NFR-MNT-01	The system shall be a modular monolith with explicit module boundaries. Modules shall not write each other's tables directly.	Must	Inspection	Exists
NFR-MNT-02	Database migrations shall be additive and checksum-locked. A committed migration shall never be edited.	Must	Inspection	Exists
NFR-MNT-03	Automated tests shall cover every invariant R1 to R9, and line coverage of the financial, approval, stock, contract and sync modules shall be at least 80 %.	Must	Test	Partial
NFR-MNT-04	The system shall be delivered with an OpenAPI specification, ER diagram, ADRs and operational runbooks, kept current with each release.	Must	Inspection	Partial
NFR-MNT-05	The system shall be containerised and deployable as multi-tenant SaaS, as a dedicated single-tenant instance, and on premises (including air-gapped), depending only on PostgreSQL and S3-compatible storage.	Must	Demo	Partial
NFR-MNT-06	The system shall emit structured logs, metrics and OpenTelemetry traces with correlation IDs, with SLO-based alerts for availability, latency, outbox lag and ledger integrity.	Must	Inspection	Partial
NFR-MNT-07	API replicas and workers shall scale horizontally. Large tenants shall be placeable on a dedicated database.	Should	Test	New
NFR-MNT-08	The supported clients shall be the current and previous major versions of Chrome, Edge, Firefox and Safari, and Android 10 or later for the mobile application.	Must	Test	Partial

7.7 Data Integrity

Invariant R1.

ID	Requirement	Pri.	Verif.	Baseline
NFR-DAT-01	Critical invariants shall be enforced by database constraints and triggers: balance, uniqueness of source events, immutability, period state, tenant identity and non-negative stock.	Must	Inspection	Exists
NFR-DAT-02	Financial, contractual and audit records shall never be hard-deleted.	Must	Test	Exists
NFR-DAT-03	Money shall use exact decimal types (never floating point), with explicit and tested rounding and allocation.	Must	Test	Exists

ID	Requirement	Pri.	Verif.	Baseline
NFR-DAT-04	The server clock shall be authoritative. Device-claimed times shall be stored as evidence beside server receipt times.	Must	Test	**Exists**
NFR-DAT-05	A legacy data migration shall reconcile opening balances, open documents and stock to the source system's totals before go-live.	Should	Test	**Partial**

8. Data Requirements

8.1 Conceptual entities

Every entity carries tenant scope, creation and change metadata, and (where relevant) status and version. Key traceability relationships are mandatory.

Entity	Key attributes	Mandatory relationships
TENANT / ENTITY / PROJECT	Currency, tax registration, calendar; project client, dates, status	Project belongs to an entity
USER / ROLE / SCOPE / DELEGATION	Principal kind, roles, project scope, limits, validity	Scopes every command
WBS / COST CODE (versioned)	Code, name, level, version, approval	Documents reference an approved version
BOQ / ESTIMATE / BUDGET (versioned)	Items, quantities, rates, revisions, transfers	Budget line traces to BOQ item
PR / RFQ / QUOTE / PO / GRN / INVOICE	Lines, cost code, quantities, prices, status	Each step references its predecessor
SUBCONTRACT / APPLICATION / CERTIFICATE	SoV, retention, advance, back-charges	Commits to budget; certificate generates payable
CONTRACT / MEASUREMENT / IPC / AR INVOICE / RECEIPT	Lines, quantities, retention, VAT base	IPC traces to measurement and contract line
STOCK ITEM / LAYER / MOVEMENT	Store, quantity, unit cost, method	Movement traces to source document; immutable
EQUIPMENT / USAGE / WORK ORDER	Hours, rate, maintenance	Usage charges a project cost code
EMPLOYEE / TIMESHEET / PAY RUN	Rates, hours, statutory components	Costs to project; journal per run
SCHEDULE / ACTIVITY / BASELINE / PROGRESS	Dates, logic, float, % complete	Progress feeds EVM
DOCUMENT / REVISION / TRANSMITTAL / RFI	Number, revision, state, recipients	One current revision
INSPECTION / NCR / PERMIT / INCIDENT	Type, location, status, actions	Linked to project and activity
JOURNAL / LINE / PERIOD / FX RATE	Source event (unique), balance, currency	Balanced; immutable once posted
PAYMENT / BANK STATEMENT / RECONCILIATION	Payee, amount, release, statement line	Settlement from statement only
ATTESTATION / AUDIT EVENT / OUTBOX EVENT	Signer key, hash chain, delivery state	Same transaction as the change

8.2 Data retention, migration and quality

- **DR-01:** Retention periods and legal holds are configurable per record type, set from confirmed legal minimums (NFR-PRV-03).
- **DR-02:** Opening balances, open documents, stock and fixed assets are loaded through validated staging and reconciled before go-live (NFR-DAT-05).

- **DR-03:** Master data, rates, tax rules and policies are effective-dated and fully audited.
- **DR-04:** Tenants can export all their data in documented open formats at any time (no lock-in).

9. Verification and Release Gates

Each invariant is tied to a test suite and recorded evidence. Failure of any critical gate blocks production approval.

Invariant	Test suite	Evidence required
R1 Ledger integrity	FINANCE: journal invariants, subledger reconciliation, journeys A-C (kernel, HTTP, browser)	Zero integrity-scan rows; reconciliation reports
R2 Separation of duties	SECURITY: forged-approval probes as tenant SQL and as schema owner; review regression suites	Before/after logs; probe results
R3 Human-only decisions	GOVERNANCE: AI/service-principal refusal tests in Go and SQL	CI results
R4 Isolation	SECURITY: cross-tenant, cross-project and export probes	Probe results; denial audit entries
R5 Exactly once	SYNC/INTEGRATION: duplicate, reordered and replayed commands, webhooks and payments	Replay test results
R6 Offline restrictions	OFFLINE: revoked device, stale base version, network cut mid-sync	Conflict evidence
R7 Governed AI	AI: prompt-injection red team, permission filtering, number verification	Red-team report
R8 Budget control	PROJECT CONTROLS: over-budget requisition/PO/subcontract, override without MFA	CI results
R9 Recoverability	OPS: PITR restore drill, encrypted off-site restore, audit-chain verification	Signed drill records

9.1 Acceptance criteria summary

- All Must requirements of the release verified with recorded evidence; no open critical or high defect.
- All applicable gates R1-R9 passed.
- Independent penetration test closed for critical and high findings (NFR-SEC-07).
- Load and soak tests meet NFR-PERF targets on a production-like environment.
- Restore test successful and documented within the last 90 days.
- UAT sign-off by finance, commercial, project controls and site operations; pilot on live projects completed.
- Tax, payroll, IFRS and data-protection mappings signed by qualified professionals.

9.2 Release plan

Release	Scope	Exit criteria
R1 Financial core + P2P (GA)	Existing baseline hardened: IAM SSO/MFA, ledger, AP/AR, treasury, P2P, inventory, contracts to cash, workflow, audit, reports, exports	Gates R1-R5, R8, R9; pentest; IdP, vault and HSM in place; signed GO
R2 Project controls + commercial	Budget revisions, forecasting, WIP and IFRS 15, subcontracts, claims, bonds, payroll, fixed assets, consolidation, tax returns	Finance and tax sign-off; payroll fixture sign-off
R3 Field + documents + QHSE	Mobile app, timesheets, equipment, document control/CDE, RFIs, submittals, quality, HSE, P6 integration	Gate R6; field pilot on two live projects
R4 Intelligence + ecosystem	Document AI, predictive risk, portals, BI connectors, report builder, load/soak at enterprise scale	Gate R7; performance targets met

10. Client Technology Position

The business requirements are independent of client technology, provided every requirement in Sections 6 and 7 is met. The positions below are recorded for agreement.

Client	Current baseline	Target
Web console	Framework-free static SPA served by the API; no authority in the client; accessibility checks automated	Keep the authority model; adopt a documented design system and component library; translation framework (ADM-04)
Mobile field app	Offline sync API and device registration exist; no native app	Offline-first app (decision D-01) with encrypted local store and durable command queue
Portals	None	Separate portal shell over the same API and authorisation rules (COL-03)
Exports	PDF, Word, Excel writers with letterhead and Ethiopic fonts	Template management (ADM-03), scheduled delivery (RPT-05)

In every case the server remains the authority for amounts, states and permissions; client code is never trusted for those decisions.

11. Assumptions and Risks

ID	Risk	Impact	Mitigation
RK-01	Scope breadth versus tier-1 suites delays core releases	Late revenue	Release plan R1-R4; MoSCoW; change control; finance and P2P first
RK-02	Statutory rules (tax, payroll, labour) misread or changed	Non-compliance, rework	Data-driven, effective-dated rules; adviser sign-off (NFR-PRV-05)
RK-03	Signer key or API process compromise	Forged approvals	Isolated signer, HSM/KMS, DB re-verification, key ceremony
RK-04	Poor site connectivity	Lost or late field data	Offline-first mobile app, idempotent sync, conflict review
RK-05	Legacy data quality at migration	Wrong opening balances	Staging, validation, reconciliation sign-off (NFR-DAT-05)
RK-06	AI misuse or prompt injection	Data leakage, wrong advice	Facts-only gateway, no decision rights, red team (R7)
RK-07	Performance at enterprise volume	Slow close, user rejection	Load tests, partitioning, dedicated DB tier (NFR-MNT-07)
RK-08	Competitors' ecosystems and integrations	Lost deals	Open API, P6/MS Project/BI connectors, migration tooling
RK-09	Dependence on missing infrastructure (IdP, vault, staging)	Release blocked	Decisions D-02 and D-03 before R1 GA

12. Quality Assurance Approach

- Requirements trace to design, code, tests and evidence in a matrix maintained for the life of the product (docs/release/FEATURE_EVIDENCE_MATRIX.csv).
- CI runs unit, integration (fresh PostgreSQL), race, security-probe, migration and browser-journey tests on every change.
- Defects are classified Critical, High, Medium or Low; no Critical or High defect may be open at release.
- Changes to financial, approval, stock, payroll, contract or sync modules require review by two engineers, and an independent review before each release.
- Gate status is generated from evidence files (tools/gate_status.py), never written by hand.

13. Commercial and Support Requirements

- **COM-01:** Support tiers with defined response times (Critical: 1 hour, 24x7 for premium).
- **COM-02:** A published roadmap, release notes, and deprecation of any API version with at least 12 months' notice.
- **COM-03:** An implementation methodology with templates (chart of accounts, workflows, roles, reports) so that a mid-size contractor can go live on finance and procurement within 12 weeks.
- **COM-04:** Training material and in-product help in every supported language.

14. Change Control

- After sign-off, every addition, deletion or modification of a requirement is raised as a written change request stating the reason, the affected IDs, and the impact on controls, compliance, schedule and cost.
- Changes to Must requirements or to invariants R1-R9 require approval by the Product Owner, the Technical Lead and the Finance/Compliance Lead.
- Approved changes increment the document version and are recorded in the revision history. Requirement IDs are never reused.

15. Agreement and Sign-off

By signing, the parties agree that this SRS is the baseline for design, development and acceptance, subject to Section 14.

Role	Name	Signature	Date
Product Owner / Client			
Technical Lead / Vendor			
Finance Lead			
Commercial / Project Controls Lead			
Operations (Site) Lead			
Security Lead			
Compliance / Legal Reviewer			
QA Lead			

Appendix A - Requirement Counts by Priority and Baseline

Section	Must	Should	Could	Total	Exists	Partial	New
6.1 Identity, Tenancy and Access Control	11	4	1	16	8	5	3
6.2 Organisation and Master Data	7	2	0	9	3	4	2
6.3 Business Development, Estimating and Tendering	3	4	1	8	2	0	6
6.4 Project Controls: Budget, Commitment, Cost and Forecast	7	3	0	10	3	3	4
6.5 Planning and Scheduling	4	4	1	9	1	1	7
6.6 Progress Measurement and Earned Value	2	3	0	5	2	0	3
6.7 Procurement and Supply Chain	5	4	0	9	4	0	5
6.8 Subcontract Management	3	4	0	7	0	0	7
6.9 Inventory and Materials	4	5	1	10	5	1	4
6.10 Equipment and Fleet	3	3	1	7	0	1	6
6.11 Workforce, Time and Payroll	4	3	1	8	0	0	8
6.12 Contracts, Revenue and Claims	6	4	0	10	3	0	7
6.13 Finance and Accounting	10	5	0	15	5	2	8
6.14 Treasury and Cash Management	2	3	1	6	1	1	4
6.15 Site Operations and Mobile	4	3	1	8	3	0	5
6.16 Document Control and BIM	5	4	0	9	0	0	9
6.17 Quality Management	3	2	0	5	0	0	5
6.18 Health, Safety and Environment	3	2	1	6	0	0	6
6.19 Workflow, Approvals and Automation	5	2	0	7	4	1	2
6.20 Reporting, Dashboards and Analytics	4	4	0	8	1	3	4
6.21 Embedded Intelligence and AI	5	3	0	8	2	4	2
6.22 Audit, Compliance and Traceability	6	1	0	7	3	3	1
6.23 Collaboration, Notifications and Portals	1	3	0	4	1	0	3
6.24 Integration and Interoperability	3	4	0	7	2	2	3
6.25 Administration, Configuration and Localisation	5	3	0	8	1	4	3
7.1 Performance and Capacity	3	3	0	6	0	0	6
7.2 Availability, Reliability and Recovery	5	1	0	6	1	3	2
7.3 Security	9	1	0	10	1	7	2
7.4 Privacy and Regulatory Compliance	5	0	0	5	0	1	4
7.5 Usability, Accessibility and Localisation	4	3	0	7	1	3	3

Section	Must	Should	Could	Total	Exists	Partial	New
7.6 Maintainability, Portability and Operations	7	1	0	8	2	5	1
7.7 Data Integrity	4	1	0	5	4	1	0
TOTAL	**152**	**92**	**9**	**253**	**63**	**55**	**135**

Baseline coverage: 63 of 253 requirements already exist (24 %), 55 are partial and 135 are new. Counts are computed from the requirement tables when this document is built.

Appendix B - Traceability: Baseline Architecture to SRS

Baseline element	Covered by SRS requirements
ADR 0001 Modular monolith	NFR-MNT-01, Section 2.1
ADR 0002/0003 Postgres invariants, exact decimal money	FIN-01..03, NFR-DAT-01..03
ADR 0005/0006 Tokens, OIDC, DB-enforced tenant identity	IAM-01, IAM-02, IAM-10
ADR 0007/0008/0015 Workflow, attestations, isolated signer	WFL-01..03, IAM-08, NFR-SEC-04
ADR 0009/0010 Subledger journals, document visibility, masked bank data	FIN-02, ORG-03, RPT-02
ADR 0011 Contract to cash	CON-01..03
ADR 0012 Stock valuation	INV-01..03
ADR 0013 CPM and EVM	SCH-01..02, EVM-01..02
ADR 0014 Distributed abuse limits	IAM-12
ADR 0016 Non-superuser schema owner	NFR-SEC-06, AUD-03
ADR 0017 Site operations and offline sync	SIT-01, SIT-02, SIT-04
Platform 0038 (settings, notifications, automation, intelligence)	ADM-03, COL-01, WFL-06, AI-01..03
docs/BACKLOG.md gap register	All requirements marked Partial or New

Appendix C - Reference Test Case Catalogue

Test ID	Expected behaviour	Requirements	Existing suite
FIN-T01	Unbalanced, closed-period or self-approved journal is refused by Go and by PostgreSQL	FIN-01..03	gate_a_ledger_test, review regressions
FIN-T02	Journeys A, B and C reconcile to the GL with zero difference	FIN-05, INV-02, AUD-06	journey_a/b/c_test, browser journeys
SOD-T01	Forged approval as tenant SQL or schema owner is refused	IAM-08, WFL-03	approval_provenance_test, attestation_*_test
SOD-T02	Concurrent decisions execute the effect exactly once	WFL-01, FIN-02	workflow_test (concurrency)
AI-T01	AI or service principal cannot approve, post, pay or release	IAM-10, AI-02	governed_ai_test
AI-T02	Injected instructions in a document cannot reach data or tools	AI-06	New: red-team suite
ISO-T01	Cross-tenant and cross-project reads and exports are refused	IAM-01, IAM-06, RPT-02	gate_a_tenancy_test, platform_test
SYN-T01	Duplicate, reordered and post-revocation sync commands	SIT-02, SIT-04	site_offline_test
BUD-T01	Over-budget PO without MFA override is refused	PCC-02, PRC-03	journey_a_controls_test
SUB-T01	Subcontract certificate retention, advance and back-charge arithmetic	SUB-02	New

Test ID	Expected behaviour	Requirements	Existing suite
PAY-T01	Payroll statutory computation against an adviser-signed fixture set	HCM-04	New
REV-T01	IFRS 15 over-time revenue and WIP schedule reconcile to the GL	PCC-06, FIN-15	New
OPS-T01	Point-in-time restore and encrypted off-site restore	NFR-AVL-02, NFR-AVL-05	BACKUP_PITR_RESTORE_
OPS-T02	Alert drill fires on induced failures only	NFR-MNT-06	metrics_drill_test
PERF-T01	Reference load and soak test	NFR-PERF-01..04	New

Appendix D - Role-Capability Matrix (initial proposal)

C = create/prepare, A = approve/certify, P = post, X = execute, R = read, - = none; * = above threshold or exception, with MFA. Project scope applies to every role except those holding tenant-wide scope. Columns group seed roles (for example, Fin./Ctrl = accountant, finance manager and controller, who are always different people for create, approve and post). To be confirmed in R1 design.

Capability	Est.	QS/Comm.	PM	Site	Proc.	Store	AP/AR	Treas.	Fin./Ctrl	CFO	HR	Aud.	Admin
BOQ / estimate	C	A	R	-	-	-	-	-	R	R	-	R	-
Budget revision / transfer	-	R	C	-	-	-	-	-	R	A	-	R	-
Requisition	-	-	A	C	R	-	-	-	-	-	-	R	-
RFQ, award, PO	-	-	R	X	C/A	-	-	-	-	A*	-	R	-
Goods receipt	-	-	R	A	R	C	R	-	R	-	-	R	-
Supplier invoice	-	-	-	-	R	-	C/A	-	A*	-	-	R	-
Payment prepare / release	-	-	-	-	-	-	-	X	A	A	-	R	-
Subcontract certificate	-	C/A	R	R	R	-	R	-	R	R	-	R	-
Measurement / IPC	-	C/A	R	A	-	-	R	-	R	R	-	R	-
Client variation / claim	-	C	R	-	-	-	-	-	R	A	-	R	-
Stock count / adjustment	-	-	-	-	-	C	-	-	A	-	-	R	-
Schedule / progress	-	R	A	C	-	-	-	-	-	R	-	R	-
Timesheets / payroll	-	-	R	C	-	-	-	-	A	R	X	R	-
Journal create/approve/post	-	-	-	-	-	-	C	-	A/P	R	-	R	-
Period close / reopen	-	-	-	-	-	-	-	-	X	A	-	R	-
Workflow / automation rules	-	-	-	-	-	-	-	-	-	A	-	R	C
Users, sessions, devices	-	-	-	-	-	-	-	-	-	-	-	R	X
Audit log	-	-	-	-	-	-	-	-	R	R	-	R	R
Assistant / risk view	-	R	R	-	R	-	-	R	R	R	-	R	-
Export reports	R	X	X	-	X	-	X	X	X	X	X	X	-

Appendix E - Open Decisions for Agreement

ID	Decision required	Suggested owner	Needed by
D-01	Mobile technology: native Android/iOS versus PWA for the offline field application (SIT-03)	Technical lead	Before R3
D-02	Hosting: SaaS region, in-country hosting partner, on-premises support policy (NFR-PRV-02)	Owner / Legal / Technical lead	Before R1 GA
D-03	Identity provider for SaaS tenants, and HSM/KMS for the approval signer (IAM-02, NFR-SEC-04)	Security lead	Before R1 GA
D-04	Payroll statutory rules and their sign-off by a qualified adviser (HCM-04)	Finance / HR / Tax adviser	Before R2
D-05	IFRS 15 method per contract type, WIP policy and JV accounting (FIN-15, FIN-08)	Finance lead / Auditor	Before R2
D-06	Tax-authority e-invoicing and fiscal obligations and format (FIN-14)	Tax adviser	Before R2
D-07	Contract templates in scope (FIDIC editions, Ethiopian PPA documents) (CON-08)	Commercial lead / Counsel	Before R2
D-08	Bank and mobile-money integrations for release 1 (INT-05, TRS-03)	Owner / Treasury	Before R2
D-09	Retention periods and legal holds (NFR-PRV-03, ADM-08)	Counsel	Before R1 GA
D-10	External AI model policy and data residency (AI-07)	Owner / Data-protection owner	Before R4
D-11	Licensing model: per user, per project value or per company tier	Owner / Commercial	Before R1 GA
D-12	Languages at launch and translation ownership (ADM-04)	Owner	Before R1 GA

End of document.